Antonie van Leeuwenhoek (2021) 114:1977–1989
https://doi.org/10.1007/s10482-021-01655-y

ORIGINAL PAPER

Draft genome sequencing and functional annotation and characterization of biofilm-producing bacterium *Bacillus novalis* PD1 isolated from rhizospheric soil

S. Ishvaria · Rajathirajan Siva Dharshini · Ranjani Manickam · K. R. Pooja · Mohandass Ramya

Received: 7 May 2021 / Accepted: 6 September 2021 / Published online: 18 September 2021

Abstract Biofilm forming bacterium *Bacillus novalis* PD1 was isolated from the rhizospheric soil of a paddy field. *B. novalis* PD1 is a Gram-positive, facultatively anaerobic, motile, slightly curved, round-ended, and spore-forming bacteria. The isolate *B. novalis* PD1 shares 98.45% similarity with *B. novalis* KB27B. *B. vireti* LMG21834 and *B. drentensis* NBRC 102,427 are the closest phylogenetic neighbours for *B. novalis* PD1. The draft genome RAST annotation showed a linear chromosome with 4,569,088 bp, encoding 6139 coding sequences, 70 transfer RNA (tRNA), and 11 ribosomal RNA (rRNA) genes. The genomic annotation of biofilm forming *B. novalis* PD1(> 3.6@OD_{595nm}) showed the presence of exopolysaccharide-forming genes (ALG, PSL, and PEL) as well as other biofilm-related genes (*comER, Spo0A, codY, sinR, TasA, sipW, degS, and degU*). Antibiotic inactivation gene clusters (ANT (6)-I, APH (3′)-I, CatA15/A16 family), efflux pumps conferring antibiotic resistance genes (BceA, BceB, MdtABC-OMF, MdtABC-TolC, and MexCD-OprJ), and secondary metabolites linked to phenazine, terpene, and beta lactone gene clusters are part of the genome.

Keywords Biofilm · Secondary metabolite · *Bacillus novalis* · Genome · Antibiotic resistance

S. Ishvaria and Rajathirajan Siva Dharshini have contributed equally to this work.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10482-021-01655-y.

S. Ishvaria · R. S. Dharshini · K. R. Pooja · M. Ramya (✉)
Molecular Genetics Laboratory, Department of Genetic Engineering, Faculty of Engineering and Technology, School of Bioengineering, SRM Institute of Science and Technology, Kattankulathur, Chengalpattu District, Tamil Nadu 603203, India
e-mail: ramya.mohandass@gmail.com

R. Manickam
SRM-DBT Platform for Advanced Life Science Technologies, SRM Institute of Science and Technology, Kattankulathur, Chengalpattu District, Tamil Nadu 603203, India

Introduction

Microorganisms in the rhizosphere are well-known for their ability to boost mineral nutrient availability, produce plant growth-stimulating chemicals, and protect plants from soil-borne pathogenic bacteria (Doornbos et al. 2012). Free-living and plant-associated bacteria are the two types of rhizospheric association that are widely found. Plant-associated bacteria can be found both inside and outside of plants. Plant roots that are in contact with rhizobacteria have a beneficial effect on plant growth and development (Backer et al. 2018). Plants benefit from these bacteria in various ways, both directly and indirectly (Etesami and Maheshwari 2018). The direct association mechanisms are modulating phytohormone levels, nitrogen fixation, iron sequestration, and resource acquisition (Tsavkelova et al. 2006; Çakmakçi et al. 2007; Kundan et al. 2015; Dewi et al. 2020). The indirect route includes the biosynthesis of the siderophores, inducing systemic resistance, antibiotics, and lytic enzymes (Dimkpa et al. 2009; Pieterse et al. 2014; Kundan et al. 2015).

Plant Growth Promoting Rhizobacteria (PGPR) acts as the biocontrol agent to eliminate soil-borne diseases (Bashan and de-Bashan 2010). Some rhizosphere organisms are linked to the plant by biofilm (Ramey et al. 2004). These biofilms could protect colonisation sites and serve as a conduit for nutrients in the rhizosphere, decreasing the amount of root exudate nutritional components available for pathogen activation or further root colonisation (Canarini et al. 2019).

Bacillus was commonly described as a dominant and plant growth-promoting bacteria among rhizospheric bacteria. *Bacillus* species can develop stress-resistant spores and release chemicals that help plants thrive and keep pathogens at bay (Radhakrishnan et al. 2017). Bacteriocins, polyglutamic acid, polylactic acid, polyhydroxyalkanoates, exopolysaccharides, and surfactants are known to be produced by them (Sansinenea 2019). Exopolysaccharides and siderophores enable plants to form biofilms and survive in hostile environments, which can aid in adaptation (Ahmad et al. 2017).

In the present study, we have isolated rhizobacteria with biofilm-forming ability from the rhizospheric soil of a paddy field. The isolate was found to be facultatively anaerobic, Gram-positive, motile, slightly curved, and round-ended rods. The optimal growth temperature of the isolate was between 30 and 40 °C, with a maximum growth temperature of 50–55 °C. The biochemical and molecular features of the isolate shares similarity with *B. novalis* (synonym *Neobacillus novalis*). *B. novalis* was initially isolated from the agricultural land of Drentse A, Netherlands by Heyrman et al. (2004). *B. vireti*, *B. soli*, *B. bataviensis*, and *B. drentensis* were among the four novel species isolated along with *B. novalis*. Patel and Gupta (2020) reclassified the Bacillus species placing *B.novalis* under *Neobacillus*.gen.nov (Niacini Clade). *B. novalis* was reported to be closely associated with *B. drentensis and B.vireti* (Lo et al. 2015). The draft genome sequence for *B.novalis* DSM 15603 ^{T} has been deposited at GenBank (LUUR00000000) without any functional annotation (Liu et al. 2016).

So far, no genome annotation and characterization were reported for *B. novalis* PD1. The current study describes the genes involved in biofilm formation, biosurfactants, secondary metabolites, and antibiotic resistance using the whole genome annotation of *B. novalis* PD1.

Materials and methods

Strain isolation and molecular identification

Rhizospheric soil was collected from the paddy field near Kattankulatur, Chennai, South India (12.8230° N, 80.0447° E). The collected soil sample was serially diluted and plated on Luria Bertani Agar (LB). A soft brown pigmented colony with an egg-shell texture surface was chosen. The genomic DNA of the selected colony was extracted from a 48 h grown culture from LB medium, using a Qiagen DNA extraction kit (Qiagen, Germany). The A260/280 ratio of isolated genomic DNA was determined using a 0.8 percent agarose gel and Nano-drop (TECAN-Infinite 200 PRO, Switzerland). The isolated genomic DNA was used for 16S rRNA amplification and whole-genome sequencing. The universal primers 27F (5′-AGAGTTTGATCCTGGCTCAG-3′) and 1492R (5′-TACGGTTACCTTGTTACGACTT-3′) were used to amplify the 16S rRNA region using polymerase chain reaction (PCR) (Chen et al. 2015). The Sanger sequencing method was used to sequence the purified PCR product (Applied Biosystems SeqStudio Genetic Analyzer, Saint Aubin, France).

16S rRNA BLAST and phylogenetic tree analysis

The Basic Local Alignment Search Tool (BLAST 2.3.0) (https://blast.ncbi.nlm.nih.gov/Blast.cgi) was used to compare the 16S rRNA nucleotide sequences to the reference sequence genome database. The top sequences from the search were selected and aligned using ClustalW. The phylogenetic tree was constructed by the neighbour joining method with a bootstrap value of 500 using MEGA- X software

(Kumar et al. 2016). *Stenotrophomonas maltophilia* ATCC 13637 T was used as an outgroup for phylogenetic tree analysis.

Genome sequencing and assembly

Whole-genome sequencing was performed with the paired-end sequencing libraries prepared using Nextera XT DNA Library Preparation kit Illumina (2×150 NextSeq-500 libraries) on an Illumina NextSeq 500 platform (Manley et al. 2016). The final library was analyzed in Bioanalyzer 2100 (Agilent Technologies, USA) using High sensitivity DNA kit as per manufacturer instructions. The Paired-End Illumina library with an insert size ranging from 700 to 1500 bp was sequenced using 2×150 cycles by SBS (Sequencing By Synthesis) chemistry in NextSeq-500. High-quality paired-end short reads of *B. novalis* PD1 raw Data obtained from Illumina NextSeq-500 as a bcl file format and converted into fastq file format using bcl2 fastq. The fastq data reads were checked for quality using FastQC v0.11.5 (Chen et al. 2018) and trimmed with trimmomatic version 0.36 (Bolger et al. 2014). Adapters and low-quality sequences were removed using Cutadapt software (Chen et al. 2018). The de novo genome assembly was assembled into scaffolds using SPAdes (Version: 3.7.1) with default parameters (–careful and –only assembler) (Bankevich et al. 2012).

Genome annotation

The assembled genome quality was verified using QUAST (Gurevich et al. 2013) and annotated using Rapid Annotation using Subsystem Technology (RAST) (http://rast.nmpdr.org) (Aziz et al. 2008), PROKKA v.2.1.1 (Seemann 2014) and Pathosystems Resource Integration Center (PATRIC) v.3.6.2 software (Gillespie et al. 2011). The genome was tested for the presence of antibiotics and secondary metabolites using antiSMASH 5.1.2 (Blin et al. 2019), and genomic islands were predicted via IslandViewer 4 (Bertelli et al. 2017). A general overview of the various biological features coded as a subsystem in the annotated genome was analyzed using PROKKA v.2.1.1 (Seemann 2014) and Rapid Annotation using Subsystems Technology (RAST) server v.2.0 (Aziz et al. 2008). The annotated genome was then analyzed using the PATRIC genome analysis server (https://www.patricbrc.org/) (Wattam et al. 2017) to identify the genome and protein features. The circular genome map was also constructed with the CG view tool.

Functional annotation

The functional prediction of protein-coding genes was accomplished by searching against clusters of orthologous groups (COG) utilizing eggNOG (Huerta-Cepas et al. 2017). The whole-genome functional annotation was done using eggNOG-mapper v2.

Comparative genomic analysis

The genome size and features of the isolate were compared with neighbouring genome sequences in the NCBI database like *Neobacillus novalis* FJAT-14,227, *B. novalis* NBRC 102,450. Further, average nucleotide (ANI) values and genomic distance of the isolate against related genomes were calculated using JSpecies (http://jspecies.ribohost.com/jspeciesws/#analyse) and through digital DNA- DNA hybridization (dDDH) tool on the DSMZ web server (http://ggdc.dsmz.de/ggdc.php#), respectively. The sequences of these genomes were annotated with the PROKKA command-line tool (Seemann 2014).

Biofilm formation assay

Biofilm formation assay was used to investigate the ability of *B. novalis* PD1 strain to form biofilms (O'Toole 2010). The overnight culture was diluted in fresh trypticase soy Broth (TSB) at a 1:100 ratio. The diluted culture (200 µL) was added to a flat-bottom 96 well microtiter polystyrene plate (Tarsons, Germany). The plates were incubated at 37 °C for 24 h and 48 h aerobically in a static condition. The experiment was carried out in triplicates utilizing different media, including trypticase soy broth, peptone water, minimal broth, and nutrient broth. After incubation, the plates were washed three times with 1X phosphate-buffered saline to remove the planktonic cells, and the wells were dried completely. The wells were stained with 200 µL of 0.01% crystal violet (CV) and incubated at room temperature for 10–15 min. After staining, the wells were washed 2–3 times with 1X PBS to remove the excess stain. Once the wells were completely dry, 200 µL of 100% ethanol was added and left at room temperature for 10–15 min to solubilize CV. After incubation, the absorbance was quantified using a UV microtiter plate reader (Thermo Scientific, Multiskan GO, US) at 595 nm (OD_{595}). Imaging of biofilm by acridine orange (AO) and CV was carried out with a sterile coverslip immersed in the suspension and incubation at 37 °C for 24 h aerobically in a static condition (Harrison et al. 2006; She et al. 2020). The swarming motility of the isolate was determined based on the colony formation on the LB agar medium (0.5% agar) (Kearns and Losick 2003).

Gas chromatography-mass spectrometry (GC–MS) analysis

Liquid–liquid extraction was performed using an equal volume of organic solvent (ethyl acetate) with the culture supernatant. The GC–MS analysis of crude extracts was performed using Agilent Technologies 7890B and 5977A. Column details are as follows: GC fused silica column, with Elite-5 MS packing (5% biphenyl 95% dimethylpolysiloxane; dimensions-30 m × 0.25 mm ID × 250 µm df). one microlitre of the sample was injected with purified helium as carrier gas at a constant flow rate of 1 mL/min at an injector temperature of 260 °C. With a 6 min long holding time, the oven temperature was maintained at 300 °C for the first 30 min and, the mass spectra were taken at 70 eV, with a scanning interval of 0.5 s. The GC–MS spectrum was analyzed by comparing it with the LIST library (Braddick 1969).

Results and discussion

Strain isolation, molecular identification, and phylogenetic tree analysis

The isolate was identified as *B. novalis* PD1 by 16S rRNA sequencing and phylogenetic analysis. The amplified fragment of 1500 bp was sequenced and subjected to BLAST analysis. Blast analysis of 16S rRNA nucleotide sequences and phylogenetic tree revealed that the strain has 98.45% similarity with *B. novalis* KB27B. Other strains like *B. vireti LMG21834* and *B. drentensis* NBRC 102,427 are also closely related to the isolate. *B. gottheilii* WCC4585 YKJ-10 and *Bacillus darkensis* strain Marseille P3515 are the distantly related species (Fig. 1). These results showed the consistency with the findings of Lo et al. (2015).

Genome sequencing, assembly, and features

The Illumina NextSeq-500 genome sequencing of the *B. novalis* PD1 strain yielded a high-quality 46,68,659 reads with 69,81,89,649 bp. The raw read was subjected to quality check and trimmed using Cutadapt software (Table 1). The trimmed data was assembled using the SPAdes assembler (Version: 3.7.1) (Figure S1). The quality values of N50 and N75 were 48,048 and 1,064, along with L50 (31) and L75 (807), respectively. The 3456 contigs cover a total length of 74,12,870 bp with the largest contig of 311,527 bp, and G + C % content is 50%. The minimum information about the novel genome, *Bacillus novalis* PD1, was represented in Table 2.

Genome annotation

The complete genome sequence of *B. novalis* PD1 was obtained with 90-fold genome coverage with a total genome size of 5,692,948 bp and GC content of 45% with 6139 coding sequences, 70 transfer RNA (tRNA), and 11 ribosomal RNA (rRNA) genes, respectively. Contigs less than 1000 bp in length were excluded from the RAST Analysis (Table 3). *B. novalis* PD1 was annotated with 1,404 hypothetical proteins and 3,342 proteins with functional assignments. The proteins with functional assignments included 1,109 with Enzyme Commission (EC) numbers, 941 with Gene Ontology (GO) assignments, and 826 with KEGG pathway mapping. Figure 2 depicts the genome circular map of *B. novalis* PD1 of the annotated genome. *B. novalis* PD1 genome contains 3,352 proteins with genus-specific protein family (PLFams) assignments and 3,566 proteins with cross-genus protein family (PGFams) assignments. In subsystem categorization, carbohydrates (690 genes) were the most abundant subsystem, followed by amino acids and derivates (640 genes), protein metabolism (324 genes), cofactors, vitamins, prosthetic groups, and pigments (279 genes), fatty acids, lipids, and isoprenoids (205 genes), RNA metabolism (194 genes), nucleosides and nucleotides (173 genes), membrane transport (170 genes), DNA metabolism (167 genes) and dormancy and sporulation (120 genes) (Fig. 3). These subsystems are required for basic cellular processes to occur. Subsystems associated with virulence, disease, and defence are found in the genome of *B. novalis* PD1.

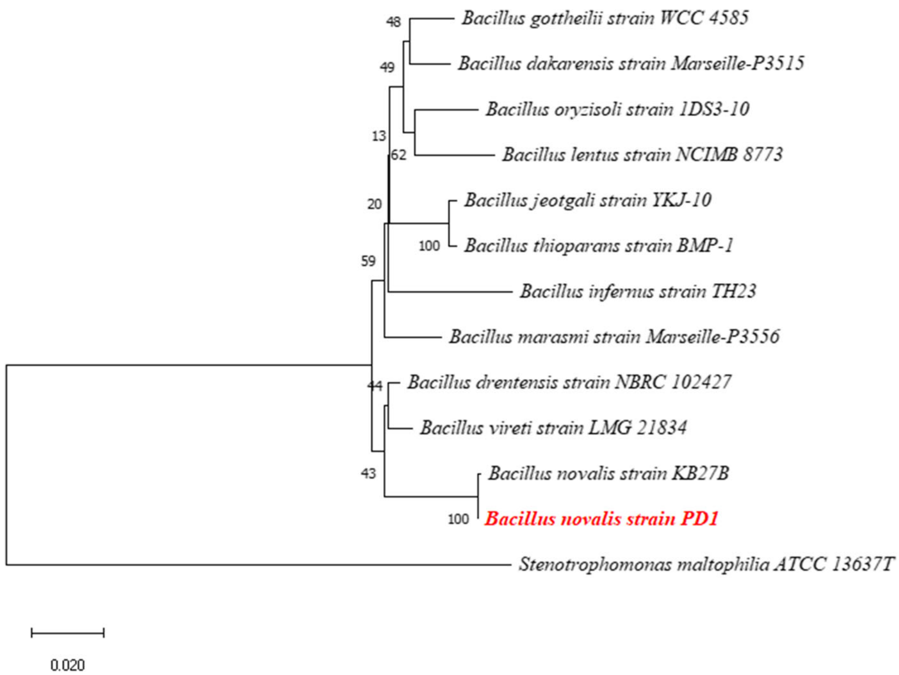

Fig. 1 Phylogenetic tree based on 16S rRNA sequence comparison highlighting the position of strain *Bacillus novalis* PD1 against other most closely related type strains

Table 1 Data of sequences before and after trimming based on the base quality check

Features	Raw Read	Trimmed Read
Paired end reads	4,668,659	4,668,659
Bases	698,189,649	698,189,649
GC%	41	41
Read length	150*2	150*2

Secondary metabolites and antimicrobial resistance analysis

The gene clusters for secondary metabolites and antibiotic resistance in *B. novalis* PD1 genome were analyzed with antiSMASH 5.1.2 (Blin et al. 2019). The genes associated with secondary metabolites are commonly involved in the biological control of phytopathogens. A total of three gene clusters for the production of secondary metabolites were predicted (Supplementary Table 1). The potential secondary metabolite-related gene clusters like phenazine (21% similarity to Streptophenazine gene cluster), terpene (6% similarity to Rhizocticin A gene cluster), and beta lactone (40% similarity with Fengycin gene cluster). Phenazine has been engaged in a lot of physiological processes, including biofilm formation and iron reduction (Hernandez et al. 2004; Letourneau et al. 2019). Terpenes have been implicated in plant growth promotion, stress alleviation, and yield increase (Piccoli and Bottini 2013). Betalactone is pivotal in inducing rapid changes in the morphology, physiology, and gene expression of roots and shoots (Schikora et al. 2016). Aside from secondary metabolites, 103 antibiotic resistance and virulence-related genes were found (Table S2). The PATRIC data source predicted approximately 47 antibiotic-resistant genes and also a significant number of virulence genes (7) (Table S3). Antibiotic inactivation enzyme encoded genes such as aminoglycoside O-Nucleotidyltransferases (ANT(6)-I), aminoglycoside phosphotransferases (APH(3')-I), and chloramphenicol resistance (CatA15/A16 family) often encode enzymes capable of inactivating particular antibiotics. The most common mechanism of chloramphenicol resistance in bacteria is acetylation via acetyltransferases or, in some cases, chloramphenicol phosphotransferases (Aakra et al. 2010). Aminoglycoside resistance is found in some strains of *Pseudomonas aeruginosa* and other gram-negative

Table 2 Taxonomic classification and general features of *Bacillus novalis* PD1 strain

Item	Description
Taxonomic Classification	
Domain	Bacteria
Phylum	Firmicutes
Class	Bacilli
Order	Caryophanales
Family	Bacillaceae
Genus	*Bacillus*
Species	*Novalis*
Strain	PD1
General features	
Shape	Round-ended rod
Cell length	0.6–1.2 m
Gram-staining	Positive
Motility	Motile
Optimal pH	7.0–9.0
Submitted to INSDC	GenBank (MW709913)
Investigation type	whole organism
Project name	*Bacillus novalis* Genome sequencing and assembly
Geographic location	India: South: Chennai
Latitude and longitude	12.8230° N, 80.0447° E
Collection date	20-01-2019
Environment (biome)	Agricultural grasslands
Environment (feature)	Rhizospheric soil
Environment (material)	Soil
Observed biotic relationship	Plant associated bacteria
Oxygen tolerance	Facultative Anaerobic
Spore description	Paracentral, slightly swollen sporangia
Type of spore	Endospore
Sequencing method	Illumina NextSeq-500
Assembly	de novo assembly using SPAdes v.3.13
Finishing strategy	Draft

bacteria due to a transport defect or membrane impermeabilization (Poole 2005). Efflux pumps conferring antibiotic resistance genes like *BceA*, *BceB*, *MdtABC-OMF*, *MdtABC-TolC*, and *MexCD-OprJ* were found. *MdtABC-TolC* in *Escherichia coli* and *Salmonella enterica* is the well-studied RND efflux pump (Nishino et al. 2006). Bacitracin efflux pumps, known as *Bce* systems, are widely found in bacteria. These genes help in intrinsic resistance in *Bacillus novalis* PD1. Antibiotic target modifying enzyme-like non-erm 23S ribosomal RNA methyltransferase (G748) helps to decrease the activity of the drug for its target.

Functional annotation

The eggNOG-mapper v2 is a tool for functional annotation of large sets of sequences based on fast orthology assignments using precomputed eggNOG v5.0 clusters and phylogenies (Huerta-Cepas et al. 2017). The collective genome annotation revealed that the isolate *B. novalis* PD1 contains CDS for metabolism, defense, and stress response, which are essential for the bacterium to survive and deal with environmental stress. The isolate has the most genes classified as unknown function, followed by amino acid transport and metabolism (Figure S1).

Table 3 RAST Annotation summary of *Bacillus novalis* PD1

Genome	*Bacillus novalis*
Domain	Bacteria
Taxonomy	Bacteria; *Bacillus novalis*
Size	5,692,948
GC Content	45
N50	86,541
L50	17
Number of Contigs (with PEGs)	936
Number of Subsystems	510
Number of Coding Sequences	6139
Number of RNAs	86

Comparative genomic analysis

The comparative genomic analysis revealed that the genome sizes and features of the compared strains were nearly identical, with only minor differences (Table 4). Remarkably low dDDH values of *B. novalis* strains indicate that they are closely related to *Neobacillus novalis* FJAT-14227 and *B. novalis* NBRC 102,450 (Lo et al. 2015). G + C content also confirms the relatedness of these organisms.

Genome analysis for biofilm formation

Biofilms are formed by bacteria as part of their survival mechanisms and are extremely common in nature. Bacterial biofilms are bacterial clusters that adhere to a surface and/or to one another while being embedded in a self-produced matrix. The biofilm matrix consists of polysaccharides (e.g., alginate),

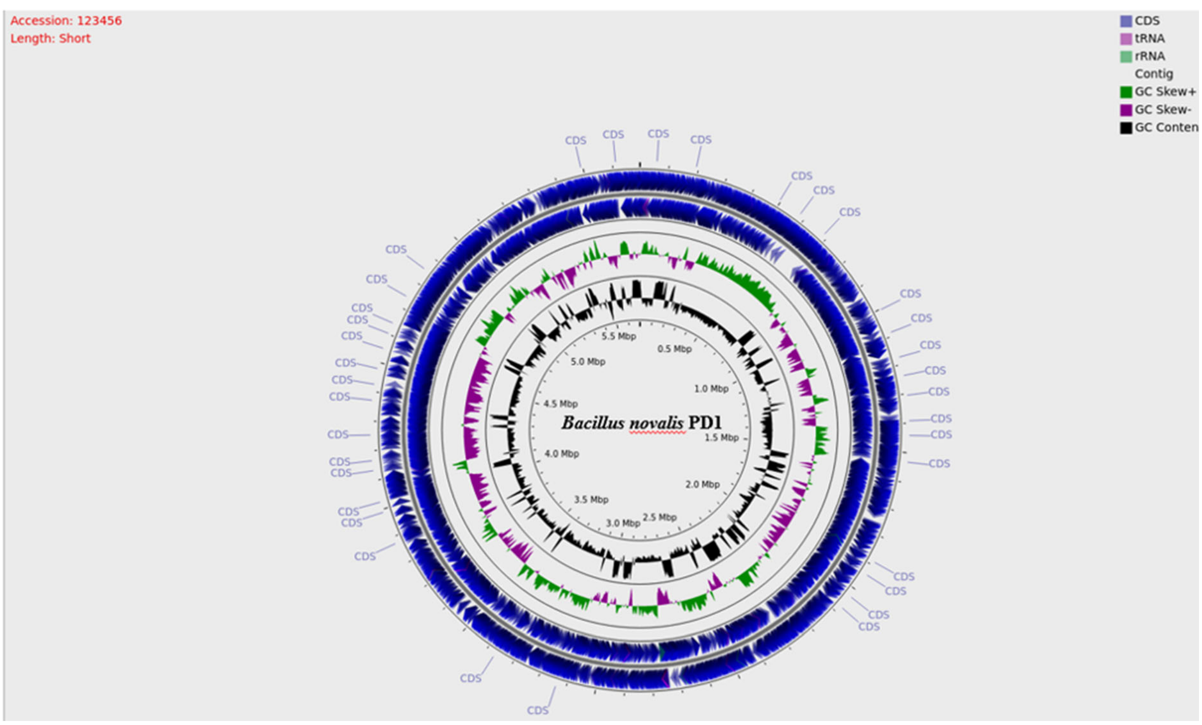

Fig. 2 Circular Genome map of *Bacillus novalis* PD1 (The overview of the genome map of *Bacillus novalis* PD1 based on 158 contigs generated via PATRIC Genome organization of *Bacillus novalis* PD1: a genome map *of Bacillus novalis* PD1 Circle representation is from Outer to inner ring—contigs (scale—x1Mbp). The blue rings represent the CDS on the forward strand, CDS on the reverse strand respectively. The grey ring shows the contig region. The ring next to grey represents RNA gene position, tRNA genes shown in pink and rRNA genes shown in green. The purple and green ring represents GC-skew. The inner ring black represents GC content

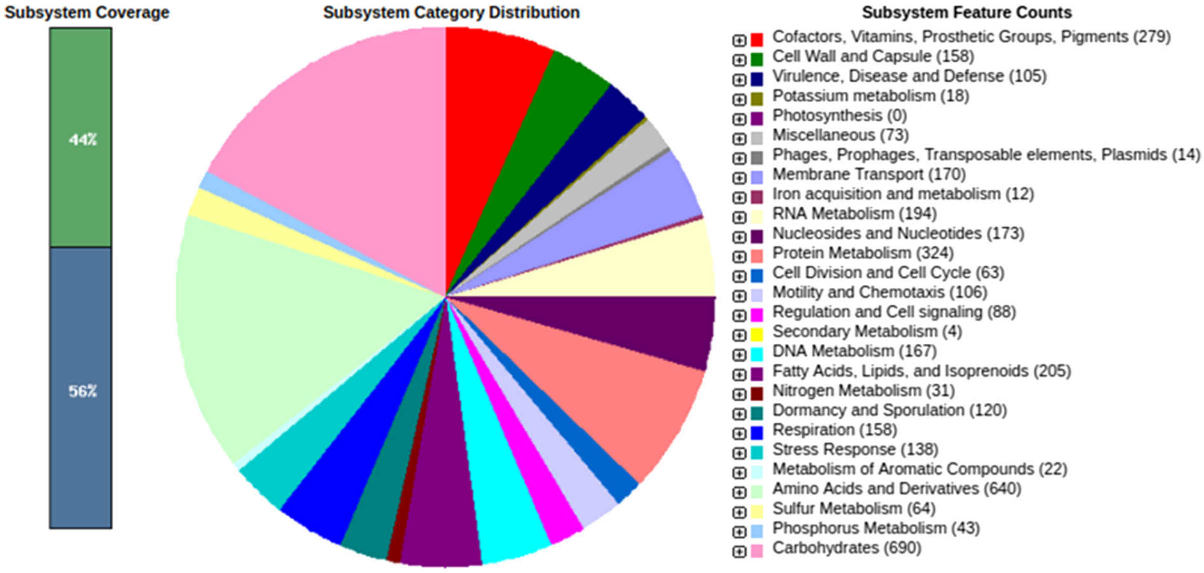

Fig. 3 Subsystem distribution of *Bacillus novalis* PD1 (RAST annotation of *Bacillus novalis* PD1: The subsystem category distribution shows the percentage distribution of the genes in different pathways, which is labelled with different colours and the number of genes in one feature is mentioned in the brackets)

Table 4 Genome comparison between *Bacillus novalis* PD1 and other neighbour species

Characteristics	*Bacillus novalis* PD1	*Neobacillus novalis* FJAT- 14,227	*Bacillus novalis* NBRC 102,450
Assembly level	Complete	Scaffold	Contigs
Genome size	5.69 Mb	5.67 Mb	5.57 Mb
Contigs	14,361	3	97
N50	86,541	5,405,962	204,086
L50	17	1	10
G + C	45	40	39.9
Genome annotation			
Genes (RNA)	10,201	5533	5445
rRNAs (5 s, 16 s, 23 s)	14	37	2
tRNAs	140	119	32
ncRNAs	2	5	5
ANIm%	NA	83.07	82.96
ANIb%	NA	74.47	74.45
dDDH	NA	20.9	20.9

NA not available

proteins (e.g., fibrin), extracellular DNA (eDNA), and lipids (Khatoon et al. 2018).

In addition to the protection offered by the matrix, bacteria in biofilms can employ several survival strategies to evade the host defense systems (Vestby et al. 2020). The matrix must be characterized by determining the relative concentration of extracellular polymeric substances (EPS) components, as well as their physicochemical characteristics and interactions (Lembre et al. 2012). The functional annotation of *B. novalis* PD1 confirms the presence of genes responsible for biofilm formation i.e., exopolysaccharides (alg, *psl, pel*) followed by genes involved in biofilm formation pathways such as *comER*, *Spo0A*, *codY*, *sinR*, *TasA*, *sipW*, *degS,* and *degU* (Table S4). *comER* plays a significant role in biofilm formation and sporulation delay in *B. subtilis* and *B. cereus*. Many studies suggested that *comER* may also be part of the regulation factor that controls activation of *Spo0A* (Yan et al. 2016). Transcription factor *Spo0A* is a master regulator for biofilm formation and sporulation. *Spo0A* regulates biofilm formation by activating a small gene *sinI*, which encodes an anti-repressor for the biofilm master repressor *SinR* (Newman et al. 2013). *SinR* directly represses two operons, *tapA-sipW-tasA,* and *epsA-O*, that are responsible for making the protein fibers (*TasA*) and exopolysaccharides (EPS) of the biofilm matrix, respectively (Elsholz et al. 2014). *DegU* is another transcription factor that is involved in regulating biofilm formation. *degU and deg S* functions as both activator and inhibitor of biofilm formation with regulation mechanism (Marlow et al. 2014). CodY is a regulator sensing the energy and the nutrient state of the bacterial cell (Hsueh et al. 2008). The regulator *CodY* was reported to be part of biofilm formation and detachment from the surface in *Bacillus cereus* (Majed et al. 2016).

LuxR homologs and complexes involved in regulating physiological processes include the production of virulence factors, biofilm formation, and secretion of extracellular enzymes (Watson et al. 2002; Skandamis and Nychas 2012). The present study found Autoinducer 2 protein (LuxS) and four transcriptional regulators (luxR) homologs in *B. novalis* PD1. LuxS is the autoinducer 2 protein that mediates quorum sensing in bacteria. LuxS converts S- ribosylhomocysteine to homocysteine and 4,5- dihydroxy- 2,3-pentanedione. 4,5- dihydroxy- 2,3 – pentanedione spontaneously cyclize to active autoinducer 2 (Van Houdt et al. 2006). LuxI/LuxR positively regulated the biofilm formation, exopolysaccharides production, and protease as co-operation (Tang et al. 2019).

Biosurfactants were associated with bacterial biofilm formation (Yamasaki et al. 2020). The existence of surfactants reduces the surface tension in the rhizosphere and enables rhizosphere rewetting (Holz et al. 2018). Surfactants promote cross-linkage in the mucilage network, which is likely to increase mucilage viscosity and cause other exudates (i.e. low molecular ones) to remain close to the root, where they are available soil microorganisms (Ahmadi et al. 2018). Biosurfactant formation is the non-ribosomal biosynthesis process catalyzed by a multienzyme complex of four modular building blocks called surfactin synthetase (Tapi et al. 2010). *B. novalis* PD1 consists of four open reading frames encoding for surfactin synthetases designated as *srf A*, *srf B*, *srf C*, and *srf D*, respectively. In addition, *sfp* genes that encode phosphopantetheinyl transferase are also found in the genome. *sfp* is the vital member of the *srfA* operon and surfactin biosynthesis gene expression (Soberón-chávez et al. 2010).

Biofilm formation assay

The quantified biofilms were scored using the method proposed by Kırmusaoğlu et al. (2019), which is as follows: $OD \leq OD_c$ no biofilm production (0), $OD_c < OD \leq 2 \times OD_c$ weak biofilm production (1 +), $2 \times OD_c < OD \leq 4 \times OD_c$—intermediate biofilm production (2 +), $4 \times OD_c < OD$—strong biofilm production (3 +). Using the criteria mentioned above, biofilm formation was observed in TSB at 48 h, while there was no biofilm formation when grown in other media supplemented (PW, MB, and NB) (Fig. 4a). The structure and profuse growth of biofilm were further imaged using fluorescent and light microscopy (Fig. 4b and c). Since the results of the MTP assay correlated with microscopic images, we confirm that *B. novalis* PD1 is a strong biofilm-forming organism.

Currently, *bacillus* species have emerged as the model organism for studying biofilm formation. In *bacillus* species, biofilm formation is usually initiated by external signals like surfactins. Bacillus species produce biofilm in the environment is parallel in the laboratory. *Bacillus* species present in the rhizosphere readily form biofilm and protect plants from pathogens (Vlamakis et al. 2013).

The isolate exhibited swarming motility which is supportive for biofilm formation (Figure. S3). Swarming motility aids in the colonisation of nutrient-rich environments as well as the acceleration of biomass production. In many cases, swarming motility requires peritrichously arranged flagella and the tightly controlled production of an extracellular matrix composed of polysaccharides, biosurfactants, peptides, and proteins (Fünfhaus et al. 2018).

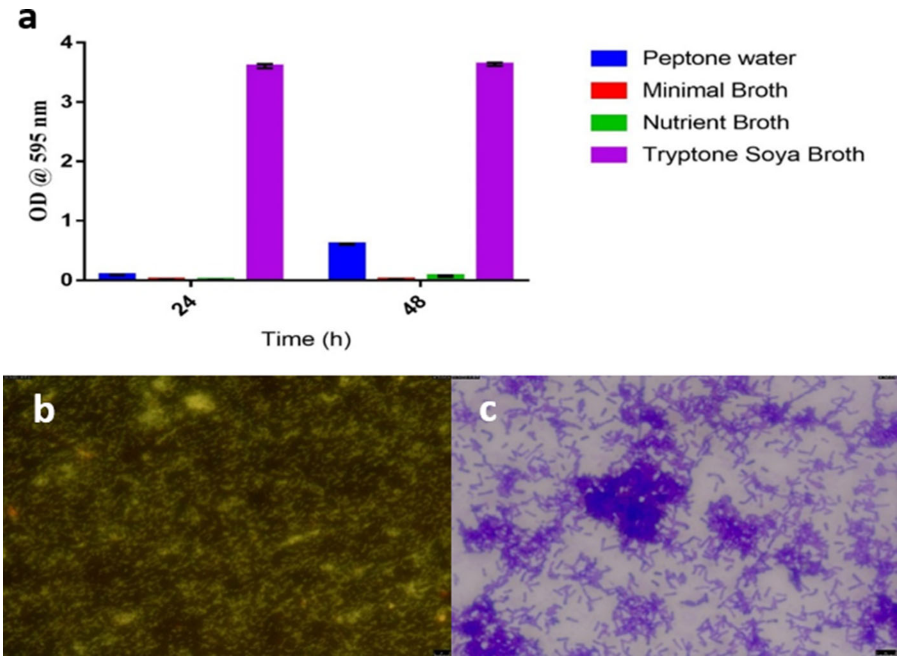

Fig. 4 **a** The spectrophotometric analysis of biofilms formation using different media (peptone water, minimal broth at different time intervals (24 h & 48 h) **b** microscopic view of the biofilm formation using acridine orange stain by fluorescent microscope and **c** Crystal violet staining of biofilm

Gas Chromatography-mass spectrometry (GC–MS) analysis

GC- MS analysis depicts the mass fragment patterns of metabolites involved in biofilm and biocontrol properties (Table- S6). Methyl stearate act as a lipid metabolism regulator, it may participate in biofilm matrix formation (Adnan et al. 2019). Pyrrolo (1,2-α) pyrazine was reported as a potential inhibitor of biofilm formation without anti-bacterial properties (Singh et al. 2019). Benzene 1,3- bis (1,1- dimethylethyl) and Phenol 2,4 -bis (1,1- dimethyl) showed the biocontrol activity, and Phenol 2,4 -bis (1,1- dimethyl) was reported as the toxic against microbes invading the rhizosphere (Mannaa and Kim 2018; Zhao et al. 2020).

Conclusion

In our study, *B. novalis* PD1 isolated from the rhizospheric soil showed biofilm formation. The genome characterization indicates genes conferring biofilm formation, antibiotic resistance, and secondary metabolites production. Our findings pave the way for the further investigation of this organism as a potential plant growth-promoting agent.

Acknowledgements We would like to acknowledge SRM-DBT Partnership Platform for Contemporary Research Services and Skill Development in Advanced Life Sciences Technologies (Order No. BT/PR12987/INF/22/205/2015), SRM Institute of Science and Technology, Chennai, Tamilnadu, India.

Author contributions SI and RSD contributed equally in this work. SI performed the experiments and drafted the manuscript. RSD performed genome analysis, genome deposition, and drafted the manuscript. RM performed genomic annotation and assembly. KRP helped in biofilm experiments and analysis. MR designed all the experiments and supervised the manuscript. MR critically reviewed and revised the manuscript.

Funding Not applicable.

Data availability The 16S rRNA sequence of *B. novalis* PD1 was submitted to GenBank with the accession number MW709913. The *B. novalis* PD1 genomic sequences obtained have been deposited in NCBI databases under the following accession numbers: PRJNA715314 (BioProject), SRR13994265 (SRA), and MW709913 (GenBank).

Declaration

Conflict of interest The authors declare that they have no conflict of interest.

References

Aakra A, Vebø H, Indahl U et al (2010) The response of Enterococcus faecalis V583 to chloramphenicol treatment the response of enterococcus faecalis V583 to chloramphenicol treatment. Int J Microbiol. https://doi.org/10.1155/2010/483048

Adnan M, Nazim Uddin Chy M, Mostafa Kamal ATM et al (2019) Investigation of the biological activities and characterization of bioactive constituents of ophiorrhiza rugosa var. prostrata (D.Don) & Mondal leaves through in vivo, in vitro, and in silico approaches. Molecules. https://doi.org/10.3390/molecules24071367

Ahmad A, Wei Y, Syed F et al (2017) Microbial Pathogenesis The effects of bacteria-nanoparticles interface on the antibacterial activity of green synthesized silver nanoparticles. Microb Pathog 102:133–142. https://doi.org/10.1016/j.micpath.2016.11.030

Ahmadi K, Razavi BS, Maharjan M et al (2018) Effects of rhizosphere wettability on microbial biomass, enzyme activities and localization. Rhizosphere 7:35–42. https://doi.org/10.1016/j.rhisph.2018.06.010

Aziz RK, Bartels D, Best A et al (2008) The RAST server: rapid annotations using subsystems technology. BMC Genom 9:1–15. https://doi.org/10.1186/1471-2164-9-75

Backer R, Rokem JS, Ilangumaran G et al (2018) Plant growth-promoting rhizobacteria: context, mechanisms of action, and roadmap to commercialization of biostimulants for sustainable agriculture. Front Plant Sci 871:1–17. https://doi.org/10.3389/fpls.2018.01473

Bankevich A, Nurk S, Antipov D et al (2012) SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. J Comput Biol 19:455–477. https://doi.org/10.1089/cmb.2012.0021

Bashan Y, de-Bashan LE (2010) How the plant growth-promoting bacterium azospirillum promotes plant growth-a critical assessment, 1st edn. Elsevier Inc, Amsterdam

Bertelli C, Laird MR, Williams KP et al (2017) IslandViewer 4: expanded prediction of genomic islands for larger-scale datasets. Nucl Acids Res 45:W30–W35. https://doi.org/10.1093/nar/gkx343

Blin K, Shaw S, Steinke K et al (2019) AntiSMASH 5.0: updates to the secondary metabolite genome mining pipeline. Nucl Acids Res 47:W81–W87. https://doi.org/10.1093/nar/gkz310

Bolger AM, Lohse M, Usadel B (2014) Trimmomatic: a flexible trimmer for Illumina sequence data. Bioinformatics 30:2114–2120. https://doi.org/10.1093/bioinformatics/btu170

BRADDICK HJJ (1969) History and Use

Çakmakçi R, Dönmez MF, Erdoğan Ü (2007) The effect of plant growth promoting rhizobacteria on Barley seedling growth, nutrient uptake, some soil properties, and bacterial counts. Turk J Agric For 31:189–199. https://doi.org/10.3906/tar-0701-8

Canarini A, Kaiser C, Merchant A et al (2019) Root exudation of primary metabolites: mechanisms and their roles in plant responses to environmental stimuli. Front Plant Sci. https://doi.org/10.3389/fpls.2019.00157

Chen J, Zhou Z, Gu JD (2015) Complex community of nitrite-dependent anaerobic methane oxidation bacteria in coastal sediments of the Mai Po wetland by PCR amplification of both 16S rRNA and pmoA genes. Appl Microbiol Biotechnol 99:1463–1473. https://doi.org/10.1007/s00253-014-6051-6

Chen S, Zhou Y, Chen Y, Gu J (2018) Fastp: an ultra-fast all-in-one FASTQ preprocessor. Bioinformatics 34:i884–i890. https://doi.org/10.1093/bioinformatics/bty560

Dewi TK, Mubarok WZ, Antonius S (2020) Study of plant growth promoting bacteria from coconut coir dust. IOP Conf Ser Earth Environ Sci. https://doi.org/10.1088/1755-1315/439/1/012037

Dimkpa CO, Merten D, Svatoš A et al (2009) Siderophores mediate reduced and increased uptake of cadmium by Streptomyces tendae F4 and sunflower (Helianthus annuus), respectively. J Appl Microbiol 107:1687–1696. https://doi.org/10.1111/j.1365-2672.2009.04355.x

Doornbos RF, Van Loon LC, Bakker PAHM (2012) Impact of root exudates and plant defense signaling on bacterial communities in the rhizosphere. A review. Agron Sustain Dev 32:227–243. https://doi.org/10.1007/s13593-011-0028-y

Elsholz AKW, Wacker SA, Losick R (2014) Self-regulation of exopolysaccharide production in Bacillus subtilis by a tyrosine kinase. Genes Dev 28:1710–1720. https://doi.org/10.1101/gad.246397.114

Etesami H, Maheshwari DK (2018) Use of plant growth promoting rhizobacteria (PGPRs) with multiple plant growth promoting traits in stress agriculture: action mechanisms and future prospects. Ecotoxicol Environ Saf 156:225–246. https://doi.org/10.1016/j.ecoenv.2018.03.013

Fünfhaus A, Göbel J, Ebeling J et al (2018) Swarming motility and biofilm formation of Paenibacillus larvae, the etiological agent of American Foulbrood of honey bees (Apis mellifera). Sci Rep 8:1–12. https://doi.org/10.1038/s41598-018-27193-8

Gillespie JJ, Wattam AR, Cammer SA et al (2011) Patric: the comprehensive bacterial bioinformatics resource with a focus on human pathogenic species. Infect Immun 79:4286–4298. https://doi.org/10.1128/IAI.00207-11

Gurevich A, Saveliev V, Vyahhi N, Tesler G (2013) QUAST: quality assessment tool for genome assemblies. Bioinformatics 29:1072–1075. https://doi.org/10.1093/bioinformatics/btt086

Harrison JJ, Ceri H, Yerly J et al (2006) The use of microscopy and three-dimensional visualization to evaluate the structure of microbial biofilms cultivated in the Calgary biofilm device. Biol Proced Online 8:194–213. https://doi.org/10.1251/bpo127

Hernandez ME, Kappler A, Newman DK (2004) Phenazines and other redox-active antibiotics promote microbial mineral reduction. Appl Environ Microbiol 70:921–928. https://doi.org/10.1128/AEM.70.2.921-928.2004

Heyrman J, Vanparys B, Logan NA et al (2004) Bacillus novalis sp. nov., Bacillus vireti sp. nov., Bacillus soli sp. nov., Bacillus bataviensis sp. nov. and Bacillus drentensis sp. nov., from the Drentse A grasslands. Int J Syst Evol Microbiol 54:47–57. https://doi.org/10.1099/ijs.0.02723-0

Holz M, Leue M, Ahmed MA et al (2018) Spatial distribution of mucilage in the rhizosphere measured with infrared spectroscopy. Front Environ Sci 6:1–7. https://doi.org/10.3389/fenvs.2018.00087

Hsueh YH, Somers EB, Wong ACL (2008) Characterization of the codY gene and its influence on biofilm formation in Bacillus cereus. Arch Microbiol 189:557–568. https://doi.org/10.1007/s00203-008-0348-8

Huerta-Cepas J, Forslund K, Coelho LP et al (2017) Fast genome-wide functional annotation through orthology assignment by eggNOG-mapper. Mol Biol Evol 34:2115–2122. https://doi.org/10.1093/molbev/msx148

Kearns DB, Losick R (2003) Swarming motility in undomesticated Bacillus subtilis. Mol Microbiol 49:581–590. https://doi.org/10.1046/j.1365-2958.2003.03584.x

Khatoon Z, McTiernan CD, Suuronen EJ et al (2018) Bacterial biofilm formation on implantable devices and approaches to its treatment and prevention. Heliyon 4:e01067. https://doi.org/10.1016/j.heliyon.2018.e01067

Kumar S, Stecher G, Tamura K (2016) MEGA7: molecular evolutionary genetics analysis version 7.0 for bigger datasets. Mol Biol Evol 33:1870–1874. https://doi.org/10.1093/molbev/msw054

Kundan R, Pant G, Jadon N, Agrawal PK (2015) Plant growth promoting rhizobacteria: mechanism and current prospective. J Fertil Pestic. https://doi.org/10.4172/2471-2728.1000155

Lembre P, Lorentz C, Di P (2012) Exopolysaccharides of the biofilm matrix: a complex biophysical world. Complex World Polysacch. https://doi.org/10.5772/51213

Letourneau MK, Marshall MJ, Grant M et al (2019) Phenazine-1-carboxylic acid-producing bacteria enhance the reactivity of iron minerals in dryland and irrigated wheat rhizospheres. Environ Sci Technol 53:14273–14284. https://doi.org/10.1021/acs.est.9b03962

Liu B, Liu GH, Zhu YJ et al (2016) Draft genome sequences of type strains Bacillus drentensis DSM 15600T and Bacillus novalis DSM 15603T. Genome Announc 4:4–5. https://doi.org/10.1128/genomeA.01423-16

Lo CI, Padhmanabhan R, Mediannikov O et al (2015) High-quality genome sequence and description of Bacillus dielmoensis strain FF4T sp. nov. Stand Genom Sci 10:1–11. https://doi.org/10.1186/s40793-015-0019-8

Majed R, Faille C, Kallassy M, Gohar M (2016) Bacillus cereus Biofilms-same, only different. Front Microbiol 7:1–16. https://doi.org/10.3389/fmicb.2016.01054

Manley LJ, Ma D, Levine SS (2016) Monitoring error rates in Illumina sequencing. J Biomol Tech 27:125–128. https://doi.org/10.7171/jbt.16-2704-002

Mannaa M, Kim KD (2018) Biocontrol activity of volatile-producing Bacillus megaterium and Pseudomonas protegens against Aspergillus and Penicillium spp. predominant in stored rice grains: Study II. Mycobiology 46:52–63. https://doi.org/10.1080/12298093.2018.1454015

Marlow VL, Porter M, Hobley L et al (2014) Phosphorylated DegU manipulates cell fate differentiation in the Bacillus subtilis biofilm. J Bacteriol 196:16–27. https://doi.org/10.1128/JB.00930-13

Newman JA, Rodrigues C, Lewis RJ (2013) Molecular basis of the activity of SinR Protein, the master regulator of biofilm formation in bacillus subtilis. J Biol Chem 288:10766–10778. https://doi.org/10.1074/jbc.M113.455592

Nishino K, Latifi T, Groisman EA (2006) Virulence and drug resistance roles of multidrug efflux systems of Salmonella enterica serovar Typhimurium. Mol Microbiol 59:126–141. https://doi.org/10.1111/j.1365-2958.2005.04940.x

O'Toole GA (2010) Microtiter dish Biofilm formation assay. J Vis Exp. https://doi.org/10.3791/2437

Patel S, Gupta RS (2020) A phylogenomic and comparative genomic framework for resolving the polyphyly of the genus bacillus: Proposal for six new genera of bacillus species, peribacillus gen. nov., cytobacillus gen. nov., mesobacillus gen. nov., neobacillus gen. nov., metabacillu. Int J Syst Evol Microbiol 70:406–438. https://doi.org/10.1099/ijsem.0.003775

Piccoli P, Bottini R (2013) Terpene production by bacteria and its involvement in plant growth promotion, stress alleviation, and yield increase. Mol Microb Ecol Rhizosph 1:335–343. https://doi.org/10.1002/9781118297674.ch31

Pieterse CMJ, Zamioudis C, Berendsen RL et al (2014) Induced systemic resistance by beneficial microbes. Annu Rev Phytopathol 52:347–375. https://doi.org/10.1146/annurev-phyto-082712-102340

Poole K (2005) Efflux-mediated antimicrobial resistance

Radhakrishnan R, Hashem A, Abd Allah EF (2017) Bacillus: A biological tool for crop improvement through bio-molecular changes in adverse environments. Front Physiol 8:1–14. https://doi.org/10.3389/fphys.2017.00667

Ramey BE, Koutsoudis M, Bodman SBV, Fuqua C (2004) Biofilm formation in plant-microbe associations. Curr Opin Microbiol 7:602–609. https://doi.org/10.1016/j.mib.2004.10.014

Sansinenea E (2019) Bacillus spp.: as plant growth-promoting bacteria. Second Metab Plant Growth Promot Rhizomicroorg Discov Appl. https://doi.org/10.1007/978-981-13-5862-3_11

Schikora A, Schenk ST, Hartmann A (2016) Beneficial effects of bacteria-plant communication based on quorum sensing molecules of the N-acyl homoserine lactone group. Plant Mol Biol 90:605–612. https://doi.org/10.1007/s11103-016-0457-8

Seemann T (2014) Prokka: rapid prokaryotic genome annotation. Bioinformatics 30:2068–2069. https://doi.org/10.1093/bioinformatics/btu153

She P, Liu Y, Wang Y et al (2020) Antibiofilm efficacy of the gold compound auranofin on dual species biofilms of Staphylococcus aureus and Candida sp. J Appl Microbiol 128:88–101. https://doi.org/10.1111/jam.14443

Singh VK, Mishra A, Jha B (2019) 3-Benzyl-Hexahydro-Pyrrolo[1,2-a]Pyrazine-1,4-dione extracted from exiguobacterium indicum showed anti-biofilm activity against pseudomonas aeruginosa by attenuating quorum sensing. Front Microbiol. https://doi.org/10.3389/fmicb.2019.01269

Skandamis PN, Nychas GJE (2012) Quorum sensing in the context of food microbiology. Appl Environ Microbiol 78:5473–5482. https://doi.org/10.1128/AEM.00468-12
Soberón-chávez G, Abdel-mawgoud AM, Hausmann R, Le F (2010) Biosurfactants and Bioengineering of Production
Tang R, Zhu J, Feng L et al (2019) Characterization of LuxI/LuxR and their regulation involved in biofilm formation and stress resistance in fish spoilers Pseudomonas fluorescens. Int J Food Microbiol 297:60–71. https://doi.org/10.1016/j.ijfoodmicro.2018.12.011
Tapi A, Chollet-Imbert M, Scherens B, Jacques P (2010) New approach for the detection of non-ribosomal peptide synthetase genes in Bacillus strains by polymerase chain reaction. Appl Microbiol Biotechnol 85:1521–1531. https://doi.org/10.1007/s00253-009-2176-4
Tsavkelova EA, Klimova SY, Cherdyntseva TA, Netrusov AI (2006) Microbial producers of plant growth stimulators and their practical use: a review. Appl Biochem Microbiol 42:117–126. https://doi.org/10.1134/S0003683806020013
Van Houdt R, Moons P, Jansen A et al (2006) Isolation and functional analysis of luxS in Serratia plymuthica RVH1. FEMS Microbiol Lett 262:201–209. https://doi.org/10.1111/j.1574-6968.2006.00391.x
Vestby LK, Grønseth T, Simm R, Nesse LL (2020) Bacterial biofilm and its role in the pathogenesis of disease. Antibiotics. https://doi.org/10.3390/antibiotics9020059
Vlamakis H, Chai Y, Beauregard P et al (2013) Sticking together: building a biofilm the Bacillus subtilis way. Nat Rev Microbiol 11:157–168. https://doi.org/10.1038/nrmicro2960
Watson WT, Minogue TD, Val DL et al (2002) Structural basis and specificity of acyl-homoserine mactone signal production in bacterial quorum sensing. Mol Cell 9:685–694. https://doi.org/10.1016/S1097-2765(02)00480-X
Wattam AR, Davis JJ, Assaf R et al (2017) Improvements to PATRIC, the all-bacterial bioinformatics database and analysis resource center. Nucleic Acids Res 45:D535–D542. https://doi.org/10.1093/nar/gkw1017
Yamasaki R, Kawano A, Yoshioka Y, Ariyoshi W (2020) Rhamnolipids and surfactin inhibit the growth or formation of oral bacterial biofilm. BMC Microbiol 20:1–11. https://doi.org/10.1186/s12866-020-02034-9
Yan F, Yu Y, Wang L et al (2016) The comER gene plays an important role in biofilm formation and sporulation in both Bacillus subtilis and Bacillus cereus. Front Microbiol 7:1–16. https://doi.org/10.3389/fmicb.2016.01025
Zhao F, Wang P, Lucardi RD et al (2020) Natural sources and bioactivities of 2,4-di-tert-butylphenol and its analogs. Toxins (Basel) 12:1–26. https://doi.org/10.3390/toxins12010035